CERN Bulletin

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http://home.cern/cern-community

CERN Council selects Mark Thomson as next Director-General, starting in 2026

Professor Thomson's five-year mandate will begin on 1 January 2026

Professor Mark Thomson selected as the new Director-General of CERN, starting in 2026. (Image: CERN)

On 6 November 2024, the CERN Council selected British physicist Mark Thomson as the Organization's next Director-General. The appointment will be formalised at the December session of the Council and Professor Thomson's five-year mandate will begin on 1 January 2026.

"Congratulations to Professor Mark Thomson on his selection as the next CERN Director-General from January 2026," said CERN's President of Council, Professor Eliezer Rabinovici. "I extend my heartfelt thanks to all the exceptional candidates. The outstanding qualities Mark Thomson displays give CERN Council assurance that he will successfully take his place in the line of visionary Directors-General who have guided CERN."

"Mark Thomson is a talented physicist with great managerial experience," said CERN Director-General Fabiola Gianotti. "I have had the opportunity to collaborate with him in several contexts over the past years and I am confident he will make an excellent Director-General. I am pleased to hand over this important role to him at the end of 2025."

Contents

"Built on long-term European collaboration, CERN is a beacon of scientific excellence and innovation, providing global leadership in research at its most fundamental level," said Professor Thomson. "CERN's mission is to unravel the mysteries of the universe, contributing to our collective pursuit of knowledge. CERN's exciting future promises groundbreaking research and discoveries that will shape our understanding of physics and, in doing so, inspire future generations of young scientists. I am honoured to become CERN's Director-General and am committed to pursuing the Organization's scientific mission, further developing technologies that will benefit society as a whole, while uniting nations in a shared commitment to advancing science for the betterment of humanity."

Professor Thomson is currently the Executive Chair of the Science and Technology Facilities Council (STFC) in the United Kingdom and a Professor of Experimental Particle Physics at the University of Cambridge. He has dedicated much of his career to CERN, where he initially contributed to precision measurements of the W and Z bosons in the 1990s, as part of the OPAL experiment at CERN's Large Electron–Positron Collider. At CERN's Large Hadron Collider (LHC), he has been a member of the ATLAS collaboration.

Since completing his doctorate in particle physics at the University of Oxford, Professor Thomson has played a significant role in advancing neutrino physics and research for future colliders. Notably, he served as co-spokesperson for the Deep Underground Neutrino Experiment (DUNE), a collaborative project led by Fermilab, which CERN supports through its neutrino platform and the construction of large cryostats to be installed deep underground in South Dakota. He has also played a pivotal role in the design and optimisation of detectors for future colliders, particularly for linear electron–positron colliders such as the International Linear Collider (ILC) and the Compact Linear Collider (CLIC).

Mark Thomson is credited in over 1000 publications and authored the widely adopted textbook *Modern Particle Physics*, used in universities globally. Beyond his research, he has held various research leadership and oversight roles at national and international level, including serving as the UK delegate to CERN's Council since 2018.

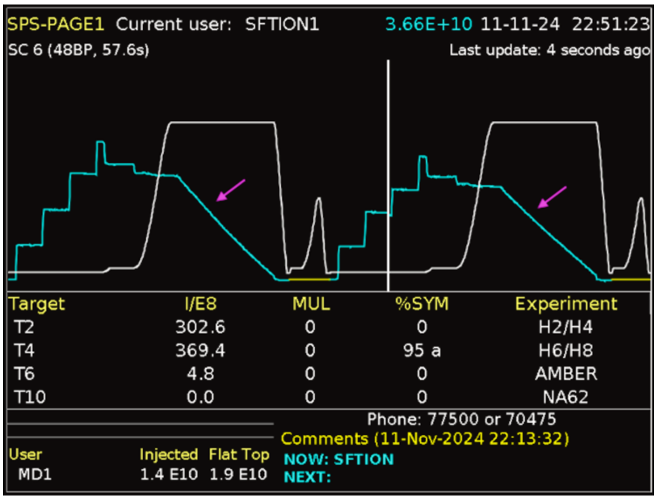
Accelerator Report: Lead ions at the heart of the accelerator complex

Lead-ion physics at the LHC and the SPS North Area is already under way and progressing well. The final lead-ion user in the PS East Area started its physics run on 13 November

The scheduled four-week lead-ion physics run began at the SPS North Area on 4 November and has since been progressing smoothly, with outstanding machine availability.

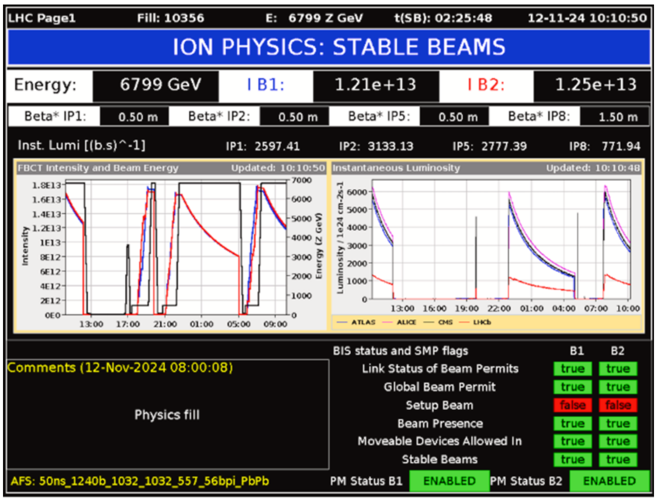
That same morning, the LHC successfully completed its proton–proton (p–p) reference run, colliding protons at 2.68 TeV to calibrate the machine ahead of the lead-ion run. Despite an electrical disturbance that caused an early beam dump at 9.51 a.m., all objectives for the p–p reference run were achieved.

Following this special run, LHC experts made some final adjustments to prepare for lead-ion operations. Among many other steps, loss maps were produced to ensure the integrity and alignment of the collimator hierarchy, confirming the machine's readiness for lead-ion collisions.



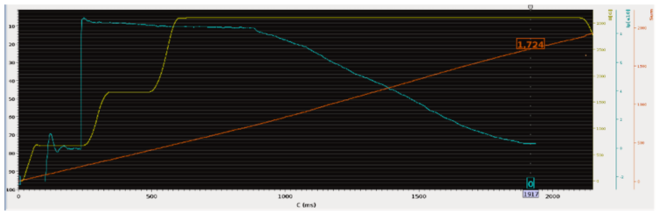
The SPS super cycle containing two lead-ion cycles destined for the SPS North Area experiments. The white line represents the electric current in the main dipole magnet, which is proportional to the energy of the beam. The blue line represents the number of lead ions in the machine. For this *beam, the SPS received four injections from the PS. The slow extraction of the lead ions to the North Area can be seen in the gradual decline of the blue line during the "flat-top" phase (see pink arrows). (Image: CERN)*

On 6 November, the LHC was filled with 119 lead-ion bunches per beam, which were then brought into collision, marking the official start of the lead-ion physics run. This initial 119-bunch configuration was the first step in a five-phase intensity ramp-up, with the aim of reaching a full machine with 1240 bunches per beam by 10 November. At each stage of this intensity ramp-up, equipment experts validated the process using detailed checklists. Each phase required at least two injections and five hours of stable beam operation. During the intensity ramp-up, all four LHC experiments conducted Van der Meer scans to calibrate their luminosity measurements, ensuring accurate data for this lead-ion run, which is scheduled to end on 25 November.



LHC page 1. In the left-hand graph, in black is the beam energy, in blue the beam intensity of the clockwise rotating lead-ion bunches, and in red the beam intensity of the counterclockwise rotating lead-ion bunches. The right-hand graph shows the instantaneous luminosity in the four experiments: blue for ATLAS, magenta for ALICE, black for CMS and red for LHCb. Note the quick decrease in intensity, which means that for the lead-ion run the LHC needs to be filled more frequently. (Image: CERN)

On 13 November, the third and final facility started its lead-ion physics run. For the next two and a half weeks, the HEARTS experiments, located in the PS East Area, will receive lead ions for their tests. HEARTS@CERN is part of the EU-funded HEARTS (High-Energy Accelerators for Radiation Testing and Shielding) project, which aims to provide access to high-energy heavy-ion radiation testing facilities for space exploitation and space exploration. This project addresses the growing need to test commercial electronic components and modules for use in space. By adjusting the energy levels, scientists and engineers can study the effects of different ion interaction depths and ionisation levels on electronic devices.


The HEARTS cycle in the PS. The yellow line represents the energy of the lead-ion beam and the blue line the number of lead ions. During the flat-top phase, the lead ions are slowly extracted toward the PS East Area, a process that takes approximately one second. (Image: CERN)

To meet the rigorous beam requirements of the HEARTS experiments, several lead-ion cycles have been carefully developed and optimised in recent weeks at the PS. The lead-ion beams produced using these cycles are slow-extracted from the PS in one-second-long spills and delivered to the T8 beam line in the East Area. With optimised transfer line optics, these beams provide uniform irradiation across sample areas of a few square centimetres. For each experiment, researchers and radiation effects engineers can choose from six different energy levels (by varying the PS energy and local degradation) and can adjust the spill intensity within three orders of magnitude, which is necessary to successfully test the large range of electronics sensitivity.

HEARTS, like the experiments in the SPS North Area, will end its physics run on 2 December.

Rende Steerenberg & Bettina Mikulec

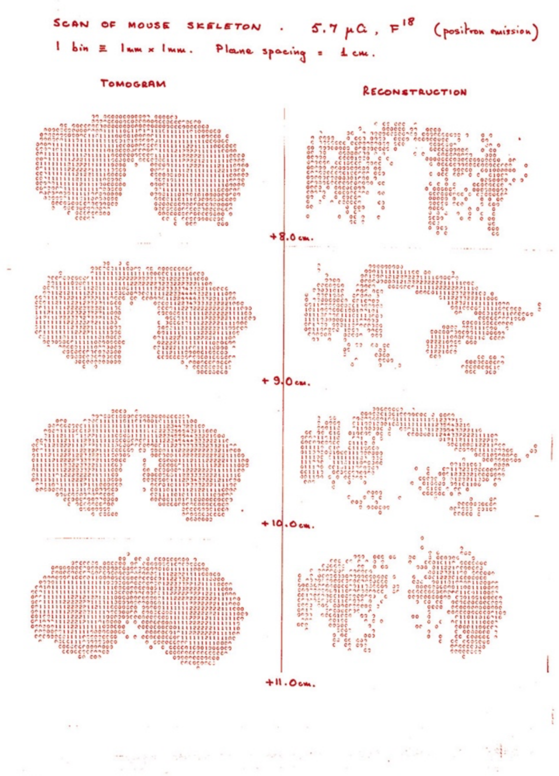
CERN70: From physics to medicine

Part 21 of the CERN70 feature series. Ugo Amaldi helped create a European network of cancer therapy centres using beams of ions, in particular carbon ions

Technology developed at CERN for accelerators and detectors has found many uses in areas beyond the field of particle physics, not least in medicine, where there are applications in both the diagnosis and treatment of disease.

Researchers at CERN have been contributing to medical research for nearly six decades, ever since radiobiology experiments began in 1965. Three years later, Georges Charpak invented the multiwire proportional chamber, a new type of detector for which he was awarded the Nobel Prize in Physics, and which came into use for medical imaging in the mid-1970s. Improvements in detectors, electronics and materials science have enabled doctors to significantly reduce the radiation dose to the patient during the examination, while improving image quality. In an important development, a prototype "positron camera" took the first PET image of a mouse in 1977, when David Townsend developed algorithms to reconstruct data from Alan Jeavons's detector using the radiobiology experiments of Marilena Streit-Bianchi. In 1979, David Townsend moved to the Geneva University Cantonal Hospital, where he helped build the hospital's first positron emission tomography (PET) scanner, based on the positron camera.

Karen Panman (left), Marilena Streit-Bianchi (centre) and Roger Paris (right) of the Radiobiology group at CERN study the effect of radiation on living cells in 1980, using an apparatus for growing broad beans that have been irradiated by 250-GeV protons. (Image: CERN)



The first mouse image taken in 1977, when positron emission tomography (PET) began at CERN. (Image: CERN)

Around the same time, physicists at CERN also became interested in using protons and carbon ions (atomic nuclei) for cancer therapy – an idea that dates back to 1946, when the use of protons and carbon ions was first proposed by the American accelerator physicist Robert Wilson, who later became the founder and first director of Fermilab. Protons and the nuclei of light elements, such as hydrogen and carbon, are better suited than X-rays to the treatment of some deep-seated cancers because their energy deposition (the "dose") can be focused to cause less damage to healthy tissue.

In the late 1970s and 1980s, CERN's accelerator experts were increasingly consulted for medical accelerator projects, in particular in the newly developing field of light ion therapy. In the 1990s, the first Medipix collaboration began and the Crystal Clear collaboration was set up to improve scintillating crystals. Both collaborations have benefited particle physics detection and medical imaging.

Then, in 1995, Ugo Amaldi, a physicist at CERN who had created the Italian TERA Foundation for radiation therapy, together with Meinhard Regler of the MedAustron project, convinced the CERN Management to start the Proton Ion Medical Machine Study (PIMMS), directed by Phil Bryant. The design of PIMMS was completed in 2000 and was later adapted by TERA to meet the needs of a centre built in Italy, CNAO (National Centre for Oncological Hadron Therapy), which began treating patients in 2011. In 2019, CERN launched the Next Ion Medical Machine Study (NIMMS) to develop cutting-edge accelerator technologies for a new generation of compact and cost-effective ion-therapy facilities.

Ongoing collaborations with medical doctors, epidemiologists and researchers are leading to new developments that help to preserve or improve our health. Examples include the role of Medipix in medical imaging, including colour X-rays, FLASH radiotherapy, the use of radioisotopes for diagnosis and treatment of diseases, machine learning and more.

Recollections

Ugo Amaldi was well known as a particle physicist at CERN when he decided to work towards the creation, in Europe, of a network of centres for cancer therapy using beams of ions, in particular carbon ions.

"My interest in medical physics dates back to the late 1950s, when I began to work at Italy's national health institute in Rome, the Istituto Superiore di Sanità (ISS), and wrote a 700-page treaty entitled Fisica delle Radiazioni. I went to ISS instead of starting an academic career because my father Edoardo, a well-known physicist and one of CERN's founders, had been a professor in Rome since 1937 and I wanted to distinguish myself from him.

At the ISS physics laboratory, I performed experiments in nuclear and particle physics and worked part-time in radiation physics and radiotherapy, a subject that I really enjoyed. From 1968 to 1973, I led a group of physicists from ISS working at CERN, and then – after a successful experiment at the Intersecting Storage Rings that discovered the increase with energy of the proton–proton cross-section – in 1973, I was offered a staff position at CERN. One thing led to another and, by 1981, I had initiated and become the spokesperson for DELPHI, one of the four LEP experiments.

By 1991, following the first results and a much-quoted paper on the supersymmetric unification of the fundamental forces, DELPHI was running very well. But it wasn't obvious to me what I should do next. I was 57 years old and I knew I wouldn't be an active experimental physicist when the LHC came online. Still, I wanted to do something new and directly useful at the frontier of what could be done with particle accelerators. I therefore decided to go back to radiation physics; a few years later, my wife Clelia said, "You went back to your first love".

Ugo Amaldi, photographed in 2023. (Image: Ugo Amaldi)

In fact, the idea of using physics to help cure people still appealed to me so, in 1991, I wrote, with Giampiero Tosi, a well-known Italian medical physicist, a proposal for a cancer therapy centre based on beams of carbon ions (and of protons), a treatment modality that, shortly after, I called "hadron therapy". One year later, we created the TERA Foundation to raise funds to cover personnel costs for the design of this centre, for which we received help from the Italian Institute for Nuclear Physics (INFN).

Treatment room of the CNAO hadron therapy centre in Italy. (Image: CNAO)

Then, as now, the main tool for cancer therapy was radiotherapy with X-rays, but dedicated proton centres were also being built. However, I was interested in some results obtained at the Lawrence Berkeley Laboratory and in papers written by Gerhard Kraft of GSI. About 1% of tumours treated with X-rays turn out to be resistant to both X-rays and protons, but it appeared that these stubborn tumours might respond if they were bombarded with the nuclei of carbon atoms. Each carbon nucleus leaves 20 times more energy than a proton in a cell and thus produces multiple, irreparable breaks of the DNA double helix. For this reason, carbon beams are effective in controlling "radioresistant tumours", such as soft-tissue sarcomas, bone sarcomas and some cancers of the brain, lung and salivary glands.

At the end of 1995, with Meinhard Regler of MedAustron, we drew the interest of the CERN Management to the design of an ion synchrotron optimised for such a medical application. This became PIMMS, the Proton Ion Medical Machine Study, on which physicists and engineers from CERN, TERA and the MedAustron project worked for four years under the direction of Phil Bryant. In parallel, with TERA, I proposed that the Italian government finance a treatment centre based on an improved version of the PIMMS synchrotron, later called the PIMMS/TERA project. As the centre would cost more than 100 million euros, this was a hard sell, but the Italian government finally put up the money in 2001. In 2003, the CNAO Foundation, with its team of 26 physicists and engineers took over the project. Sandro Rossi, who had been the TERA Technical Director for almost ten years, became the CNAO Technical Director and, later, its Director-General. With the help of INFN, in autumn 2005, the construction of the Centre started in Pavia, an ancient university town close to Milan. In 2011, the first proton irradiation took place and, in 2012, exactly 20 years after the creation of TERA Foundation, the first patient was treated with a beam of carbon ions.

When the first patients were treated at CNAO, I felt that I'd done very interesting things in physics, but hadron therapy was the best thing I'd done in my professional life.

I am now President Emeritus of CNAO, and I have the satisfaction of seeing that the Centre is still growing. More than 5000 patients have been treated, 55% with carbon ions, and more and more patients are children: in 2023, 62 paediatric tumours were irradiated. Moreover, a proton gantry is being installed and a novel boron neutron capture therapy system will be operative in 2026. Overall, one can say that CNAO, as a spin-off of CERN and TERA, fully justifies my motto "Physics is beautiful and useful".

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This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary, and was updated with the help of Ugo Amaldi in 2024. Ugo was one of the speakers at the recent CERN70 public event "From particle physics to medicine". A new PET digital learning module is available now for high-school students to help save a virtual patient, find out more via cern.ch/petlearningmodule.

CMS develops new AI algorithm to detect anomalies

During LHC Run 3, researchers at the experiment have deployed an innovative machine learning technique that will improve the data quality of one of the detector's most crucial components

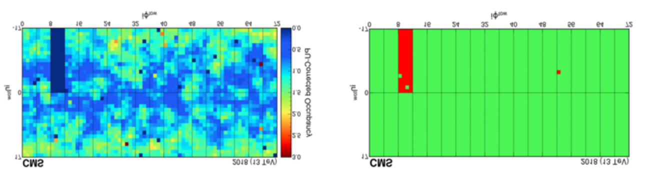
In the quest to uncover the fundamental particles and forces of nature, one of the critical challenges facing high-energy experiments at the Large Hadron Collider (LHC) is ensuring the quality of the vast amounts of data collected. To do this, data quality monitoring systems are in place for the various subdetectors of an experiment and they play an important role in checking the accuracy of the data.

One such subdetector is the CMS electromagnetic calorimeter (ECAL), a crucial component of the CMS detector. The ECAL measures the energy of particles, mainly electrons and photons, produced in collisions at the LHC, allowing physicists to reconstruct particle decays. Ensuring the accuracy and reliability of data recorded in the ECAL is paramount for the successful operation of the experiment.

During Run 3 of the LHC, which is currently ongoing, CMS researchers have developed and deployed an innovative machine-learning technique to enhance the current data quality monitoring system of the ECAL. Detailed in a recent publication, this new approach promises to make the detection of data anomalies more accurate and efficient. Such real-time capability is essential in the fast-paced LHC environment for quick detection and correction of detector issues, which in turn improves the overall quality of the data. The new system was deployed in the barrel of the ECAL in 2022 and in the endcaps in 2023.

The traditional CMS data quality monitoring system consists of conventional software that relies on a combination of predefined rules, thresholds and manual inspections to alert the team in the control room to potential detector issues. This approach involves setting specific criteria for what constitutes normal data behaviour and flagging deviations. While effective, these methods can potentially miss subtle or unexpected anomalies that don't fit predefined patterns.

In contrast, the new machine-learning-based system is able to detect these anomalies, complementing the traditional data quality monitoring system. It is trained to recognise the normal detector behaviour from existing good data and to detect any deviations. The cornerstone of this approach is an autoencoder-based anomaly detection system. Autoencoders, a specialised type of neural network, are designed for unsupervised learning tasks.



An image from ECAL data with anomalous regions (left) which, when passed through the machine-learning system, produces the easily identifiable colour map on the right, showing anomalous regions in red and good regions in green. (Image: CMS experiment)

The system, fed with ECAL data in the form of 2D images, is also adept at spotting anomalies that evolve over time thanks to novel correction strategies. This aspect is crucial for recognising patterns that may not be immediately apparent but develop gradually.

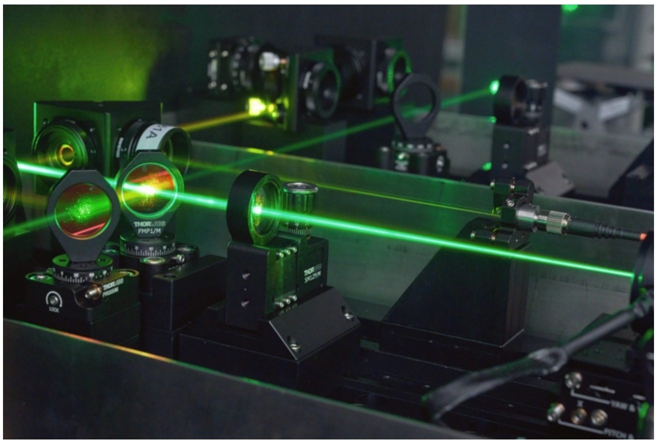
The novel autoencoder-based system not only boosts the performance of the CMS detector but also serves as a model for real-time anomaly detection across various fields, highlighting the transformative potential of artificial intelligence. For example, industries that manage large-scale, high-speed data streams, such as the finance, cybersecurity and healthcare industries, could benefit from similar machine-learning-based systems for anomaly detection, enhancing their operational efficiency and reliability.

CMS is just one of many experiments at CERN that is improving its performance using AI, automation and machine learning. Read more about this here.

CMS collaboration

Celebrating one year of CERN Venture Connect

A year into its journey, CERN Venture Connect has become a launchpad for deep-tech startups



One of the laser technologies to which startups participating in the CERN Venture Connect Programme benefit from. (Image: CERN)

One year ago, CERN launched the CERN Venture Connect (CVC) Programme, a groundbreaking entrepreneurship initiative designed to empower deep-tech ventures across the Laboratory's Member and Associate Member States at every stage of their journey, from seed to late-stage growth. Focused on accelerating startups by giving them access to state-of-the-art technologies and to a global support network, CVC has quickly grown into a thriving hub for innovation.

As showcased in the CERN Venture Connect Summit taking place today, in its first year, CVC has built a dynamic ecosystem of startups, investors, mentors and key players within the deep-tech sector, actively supporting three pioneering startups, establishing partnerships with over 40 organisations across 15 countries and providing fast-track access to seven cutting-edge CERN technologies.

"Startup talent is everywhere, but access to the right network and resources isn't always available locally," says head of the Programme Linn Kretzschmar. "CERN Venture Connect is designed to bridge that gap, connecting founders with world-class technologies and a global network of investors and mentors. Our fast-tracked licensing agreements and access to technology prototypes enable startups to accelerate their growth and create lasting societal impact."

Startups joining CVC gain access to groundbreaking CERN technologies that were originally developed for fundamental physics research and the operation of the world's largest particle accelerator infrastructure but are now primed for real-world applications. These technologies, such as advanced laser systems, cooling systems and time-synchronisation tools, aim to revolutionise fields such as quantum computing, healthcare, telecommunications, finance and smart grids.

One notable example is the structured laser beam system, which was originally designed for aligning accelerator components at CERN. This technology is now being used by the Dutch startup InPhocal to mark products in the food and beverages industries, outperforming traditional approaches in terms of speed and cost. Moreover, it does not require the use of toxic printing inks and has the capability to print on curved surfaces.

In addition, CVC's comprehensive support network helps startups secure funding, build expertise and form strategic partnerships. "The level of interest from our partners has been outstanding," says Chris Hartley, head of CERN's Industry, Procurement and Knowledge Transfer department. "We are excited to collaborate with incubators, investors and industry experts who share our vision of supporting founders on their entrepreneurial journey. The practical application of CERN's technologies for societal benefit is of paramount importance to us."

CERN Knowledge Transfer group

Find out more, including guidelines on how to apply and a list of the Programme's partners, on the CVC website.

CERN IdeaSquare turns ten

On 8 November, the IdeaSquare community gathered to celebrate ten years of the Innovation Space at CERN

The celebration, held at CERN Globe of Science and Innovation, included a series of engaging talks about the past and future of IdeaSquare. (Image: Daniel Dobos)

CERN IdeaSquare celebrated its tenth anniversary on 8 November 2024. The celebration, open to the CERN community and significant people from IdeaSquare's history, invited attendees to explore the decade-long journey of the innovation space, as well as look forward to its exciting future.

Since 2014, this vibrant hub for science and innovation has provided unique facilities for rapid prototyping, education programmes, workshops and hackathons. It is used by the CERN collaborations, CERN Knowledge Transfer projects and EU initiatives linked to the UN sustainable development goals.

"It's so simple, it's difficult to understand," said Chris Hartley, the head of CERN's IPT department, in his opening speech.

Want to find out more about how the CERN Community can use IdeaSquare? Visit this link or check out the IdeaSquare website. Photos of the event can be found here.

Naomi Dinmore

Celebrating 25 years of service to CERN in 2024

Staff members who marked 25 years of service to CERN in 2024 were invited by the Director-General to the traditional ceremony held in their honour on 4 November 2024.

The photos from the ceremony and the list of the 63 staff members concerned can be viewed in this album (restricted access).

We thank them all warmly for their commitment and wish them continued success at CERN!

HR department

Computer Security: Security Rules revised

With the 2023 cybersecurity audit and the subsequent approval by CERN's Extended Directorate of the new Cybersecurity Policy, which parallels, complements and clarifies the CERN Computing Rules aka Operational Circular 5 (OC5), the foundations have been laid to review, augment and produce new Subsidiary Rules to better explain "what goes and what does not" with regards to CERN's computing facilities and the computing resources they serve. Let's look at the full package and how it impacts you.

For one, the **Computing Rules and the Cybersecurity Policy** require that anyone using or contributing to CERN's computing facilities (e.g. its network, CERN-owned devices, on-site or cloud-based computing services) actively contribute to the implementation of the Rules and Policy through exemplary conduct – namely by:

- acting in compliance with the Rules, including the Subsidiary Rules;
- actively seeking information to minimise risks;

- avoiding dangerous situations for their equipment and CERN's computing facilities; and
- assuming the responsibilities assigned to them.

As such, and unless responsibility is delegated when using central services, **all owners of computing resources connected to or provided to them by CERN's computing facilities are ultimately responsible for the compliance of their actions and their resources with these Rules**.

For two, in order to help all users of CERN's computing facilities to better fulfil their responsibilities and to better understand "what goes and what does not", the aforementioned set of dedicated "Subsidiary Rules" provide managerial and technical guidance on how to use CERN's computing facilities in a secure fashion. Like OC5 and the Cybersecurity Policy, these Subsidiary Rules are binding (see OC5 II 8a). Any derogation from these Rules requires written approval by CERN's Computer Security Officer and may be entered in the CERN/IT Risk Register. Non-compliance with any of these Rules might lead to sanctions, e.g. reduced functionality (limited connectivity, e.g. "throttling"), the termination of service ("blocking") or administrative measures (see OC5 V).

Subsidiary Rules, whether newly created or to be updated, are discussed and approved (or rejected) in the just-established Computer Security Board, comprised of appointed Computer Security Liaisons as representatives of CERN sectors/departments/units and the experiments. In the next couple of months, this Board will review all the current Subsidiary Rules and, in order to complement the implementation of the cybersecurity audit's recommendations, create additional Rules.

Computer security revised – but in the end, however, there shouldn't be any surprises for you: the same rules as always (OC5 dates back to the year 2000!) apply to you and your use of CERN's computing facilities. Now they are just clearer, more concrete and more explicit. A big thank you to you all for helping to secure and protect CERN!

Computer Security Office

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: https://cern.ch/official-news.

"Twilight burglary" prevention campaign

The Swiss authorities warn of high risks of burglary at this time of year. Risks can be considerably reduced by taking simple preventive measures:

- Simulate your presence by using a light on a timer, for example
- Keep valuables out of sight and systematically lock doors and windows
- Inform your neighbours if you are going to be away for a long time and have your letterbox checked
- If you notice any suspicious behaviour, including that of your neighbours, call 117 immediately (in France, call 17)

More information on the Swiss Crime Prevention website at: https://www.skppsc.ch/fr/sujets/cambriolage

In the event of an incident, please refer to the information on the following websites:

- Switzerland: https: //www.ge.ch/vol-cambriolage-dommage-propriete/cambriolages

- France: https://www.gendarmerie.interieur.gouv.fr/conseils/habitation/cambriolage-comment-reagir-et-limiter-les-risques

Host State Relations service
Relations.secretariat@cern.ch Tel.: 75152

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

26 Oct 2024 – 2 Mar 2025 | Knowledge Sharing | Globe of Science & Innovation | CERN & Muséum d'histoire naturelle de Genève | EN & FR
Curious connections exhibition / Exposition Dialogues Insolites

4–15 Nov | Knowledge Sharing | Main Building Exhibition | CERN & Society Foundation | EN
CERN & Society Foundation: 10 years bridging science and society

14 Nov | Knowledge Sharing | CERN | Education locale | FR
Futur en tous genres

16 Nov 8:30 | Knowledge Sharing | UniMail, Geneva | Education locale | FR
Élargis tes Horizons: Journée ateliers scientifiques et technologiques pour les jeunes filles

18 Nov 14:00 | At CERN | Online | Human Resources | FR
MERIT 2025: Réunion publique en français

19 Nov 14:00 | At CERN | Main Auditorium | LGBTQ+@CERN | EN
LGBTQ+ in STEM Day – Colloquium and Cupcakes

19 Nov 19.30 | At CERN | CERN Science Gateway | Public event | FR
Danser avec l'Évolution

21 Nov 14:00 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
The Voice of Science – Meet Dr Claire Malone

21 Nov 16:30 | Knowledge Sharing | Council Chamber | CERN Colloquium | EN
Is dark energy weakening?

26 Nov 13:00 | Knowledge Sharing | Online | KT Seminars **| EN**
CERN Workshop on Global Health: Presentation of Results Webinar

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN
DRD1 Gaseous Detectors School

29 Nov–1 Dec | Knowledge Sharing | CERN et Hepia | CERN et Hepia | FR
Les journées Codez la science 2024

6 Dec 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
Meet the authors of "The stars' last dance" ("La dernière danse des étoiles")

11 Dec 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN
Meet the authors of « Les Aventures Extraordinaires des Globifiques »

Registration now open

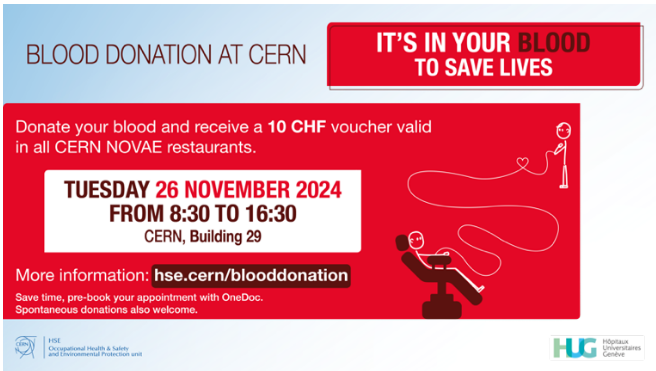
10–14 Mar | Physics | Ferney-Voltaire | CERN Accelerator School | EN
Basics of Accelerator Physics and Technology

30 Apr – 13 May | Physics | Costa Rica | CLASHEP | EN
2025 CERN Latin-American School of High-Energy Physics

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events
Indico also shows ALL events happening today, this week and in a calendar view.
If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Next blood donation campaign on 26 November

Contribute to this essential act of solidarity that benefits us all



Two blood donation campaigns have been run so far this year, in March and July, enabling the collection of some 200 donations in total, with 50-60 new donors coming to donate each time. This is a great result, earning CERN the 2nd place as the region's biggest blood donor after the *Hôpitaux universitaires de Genève* (HUG). Thank you to all donors!

Whether we are regular or new donors, our contributions are crucial to ensure sufficient blood availability for those in need in the area. The Geneva blood barometer shows that blood stocks remain extremely low, notably for blood groups AB-, O-, A- and B-. Therefore, the CERN Medical Service will join forces once again with the HUG for a third blood donation session on **Tuesday 26 November, from 8.30 a.m. to 4.30 p.m., in Building 29**.
Can we still improve on past donations and collect more?
As a thank you, the HUG will be giving each donor a 10 CHF voucher, to be used in Novae's Restaurants 1 and 2 at CERN.

Before coming to the Building 29 donation point, please make sure that:

- You are in good health and have no symptoms such as fever, cough, a cold or breathing difficulties;
- You are eligible to give blood:
  - Complete the brief online pre-donation questionnaire (in French)
  - Please consult this information sheet and this questionnaire issued by the HUG.
  - Note that only the pre-donation conversation with the nurse or doctor on the day can confirm your eligibility).
  - If returning from travel abroad or arriving from another country, perform the travelcheck on the HUG website (in French)
  - If you have recently been vaccinated against the flu or COVID, ensure 48 hours have passed since receiving the vaccination.
  - Please bring proof of ID with you on the day of the donation.

Save yourself time: pre-book your appointment with OneDoc
The possibility of pre-booking your appointment via the OneDoc platform is made available on this link. Spontaneous donors will be able to come at their convenience without an appointment, mindful that there may be some waiting time on the day.
We hope to see many of you at the donation point!

HSE unit

Transformations at Geneva Cornavin Station

The Swiss Mission was informed by the Swiss railways (CFF/SBB) of modernisation works planned at Geneva Cornavin station, with major impacts on rail and road traffic from Thursday 21 to Monday 25 November, including:

- A total interruption of rail traffic for 20 hours from 7pm on Saturday 23 November to 3pm on Sunday 24 November (see perimeter under this link).
- A total interruption of SNCF TER traffic between Geneva and Bellegarde and a reduction on

the *Léman Express* from 7.50pm on Friday 22 November to 4.30am on Monday 25 November.

For further details, including the alternative arrangements, customers are asked to check the information available on the CFF website and on the CFF Mobile app.

Host State Relations service
Relations.secretariat@cern.ch Tel.: 75152

Help us drive change – share your mobility habits by taking this short survey!

CERN's sustainability efforts rely on insights from our community, and understanding our collective mobility habits is key

CERN is committed to providing optimal mobility solutions for its entire community, facilitating access to and movement around the sites. The SCE department is responsible for providing and maintaining the infrastructure and services needed to enable people to move around the CERN sites easily. A dedicated working group comprising representatives from all sectors of the Organization meets regularly to support SCE in its continuous efforts to improve mobility at CERN, ensuring variety and flexibility of soft mobility options and providing optimal connectivity for all road users, whatever their mode of transport. In the wider sustainability context, all our actions ultimately contribute to minimising the greenhouse gas (GHG) emissions arising from mobility at CERN.

Your responses to this short, seven-question survey will support continuous improvements to our mobility infrastructure and also help to calculate CERN's Scope-3 GHG emissions arising from mobility for the next Environment Report.

The survey takes only a minute or so to complete, and will be open until Friday, 6 December and we thank you in advance for your time and contributions.

Every response matters – take the survey* today! https://cern.ch/mobility-survey

SCE department and HSE unit

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Find out more about CERN's extensive mobility solutions on this webpage: https://sce-dep.web.cern.ch/mobility-plan, which also includes the results of the previous mobility survey (2023).

Past information about GHG emissions at CERN can be found in the Emissions chapter of the third Environment Report (2021–2022).

** The survey is anonymous and has been developed in Microsoft Forms. A CERN login is required. The associated privacy notice can be consulted here.*

Ethics at CERN: take the new online course

At the beginning of the year, in order to make our ethics-related framework more easily accessible, the HR department grouped all the relevant information and resources under a single point of reference at https://hr.web.cern.ch/ethics.

To go further and ensure that every CERN staff member is aware of the guidelines of our ethical framework, the HR department has developed an e-learning course titled "Ethics at CERN", now available in English and French on the CERN Learning Hub.

Ethics is essential, in particular in our workplace. It contributes to a positive and productive work environment whilst fostering trust among us, as CERN contributors, and also the trust of our stakeholders, such as Member States, contractors, and research institutions.

We encourage all CERN staff to take this new e-learning course. At CERN, our ethics-related framework aims to guide us to act in accordance with the Organization's values and the conduct expected of us, and to establish and maintain a respectful working environment. It is up to each and every one of us to respect it.

For more information or questions about ethics at CERN, please contact info-codeofconduct@cern.ch.

HR department

Obituaries

Ian Shipsey (1959 – 2024)

(Image: University of Oxford)

Experimental particle physicist Ian Shipsey, a remarkable leader and individual, passed away suddenly and unexpectedly in Oxford on 7 October.

Ian was educated at Queen Mary University of London and the University of Edinburgh, where he earned his PhD in 1986 for his work on the NA31 experiment at CERN. Moving to the US, he joined Syracuse as a post-doc and then became a faculty member at Purdue, where, in 2007, he was elected Julian Schwinger Distinguished Professor of Physics. In 2013 he was appointed the Henry Moseley Centenary Professor of Experimental Physics at the University of Oxford.

Ian was a central figure behind the success of the CLEO experiment at Cornell, which was for many years the world's pre-eminent detector in flavour physics. He led many analyses, most notably in semi-leptonic decays, from which he measured four different CKM matrix elements, and oversaw the construction of the silicon vertex detector for the CLEO III phase of the experiment. He served as co-spokesperson between 2001 and 2004 and was one of the intellectual leaders that saw the opportunity to re-configure the detector and the CESR accelerator as a facility for the precise exploration of physics at the charm threshold. The resulting CLEO-c programme yielded many important measurements in the charm system and enabled critical experimental validations of lattice-QCD predictions.

Within the CMS collaboration, Ian played a leading role in the construction of the forward-pixel detector, exploiting the silicon laboratory he had established at Purdue. His contributions to CMS physics analyses were no less significant. These included the observation of upsilon suppression in heavy-ion collisions (a smoking gun for the production of quark–gluon plasma) and the discovery, reported in a joint Nature paper with the LHCb collaboration, of the ultra-rare decay $B_s \rightarrow \mu^+\mu^-$. He was also an influential voice as CMS Collaboration Board chair (2013–2014).

After moving to the University of Oxford and, in 2015, joining the ATLAS collaboration, Ian became Oxford's ATLAS team leader and established state-of-the-art clean rooms, which are used for the construction of the future inner tracker (ITk) pixel end-cap modules. Together with his students, he contributed to measurements of the Higgs boson mass and width, and to the search for its rare di-muon decay. Ian also led the UK's involvement in LSST (now the Vera Rubin Observatory), where Oxford is providing deep expertise for the CCD cameras.

Following his tenure as the dynamic head of the particle physics sub-department, Ian was elected head of Oxford Physics in 2018 and re-elected in 2023. Among his many successful initiatives, he played a leading role in establishing the UKRI "Quantum Technologies for Fundamental Physics" programme, which is advancing quantum-based applications across various areas of physics. With the support of this programme, he led the development of novel atom interferometers for light dark matter searches and gravitational wave detection.

Ian took a central role in establishing roadmaps for detector R&D, both in the US and (via ECFA) in Europe. He was one of the coordinators of the ECFA R&D Roadmap Panel and a driving force behind it, as well as being co-chair of the US effort to define the basic research needs in this area. As chair of the ICFA Instrumentation, Innovation and Development Panel, he promoted R&D in instrumentation for particle physics and the recognition of excellence in this field.

Among his many prestigious honours, Ian was elected a Fellow of the Royal Society in 2022 and received the James Chadwick Medal and Prize from the Institute of Physics in 2019. He served on numerous collaboration boards, panels and advisory and decision-making committees shaping national and international science strategies.

The success of Ian's career is even more remarkable given that he lost his hearing in 1989. He received a cochlear implant, which restored limited auditory ability. He gave unforgettable talks on this subject, explaining the technology and its impact on his life.

Ian was an outstanding physicist and also a remarkable individual. His legacy is not only an extensive body of transformative scientific results, but also the impact that he had on all who met him. He was equally charming whether speaking to graduate students or lab directors. Everyone felt better after talking to Ian. His success derived from a remarkable combination of optimism and limitless energy. Once he had identified the correct course of action, he would not allow himself to be dissuaded by cautious pessimists who worried about the challenges ahead. His colleagues and many graduate students will continue to benefit for many years from the projects he initiated. The example he set as a physicist, and the memories he leaves as a friend, will endure still longer.

Many of Ian's scientific successes were achieved in professional collaboration with his wife, Daniela Bortoletto, who survives him, together with their daughter, Francesca.

His friends and colleagues

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This obituary will also appear in the CERN Courier*. Read the interview with Ian Shipsey published in the* CERN Courier *in 2020.*